

Answer **all** questions.

1. The nutrition information in **Table 1.1** is taken from a pack of dried pasta.



Table 1.1

TYPICAL VALUES	Per 100 g (of dried pasta)
Energy	761 kJ
Fat	1.3 g
of which – saturated	0.1 g
of which – unsaturated	 g
Carbohydrates	25.0 g
of which – sugars	2.3 g
Fibre	7.7 g
Protein	13.0 g
Salt	0.05 g



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(a) (i) Calculate the value for unsaturated fats. **Write your answer in Table 1.1.** [1]
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(ii) State the name of the nutrient which makes up most of the carbohydrates in the dried pasta. [1]

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(iii) State the importance of a low-salt diet. [1]

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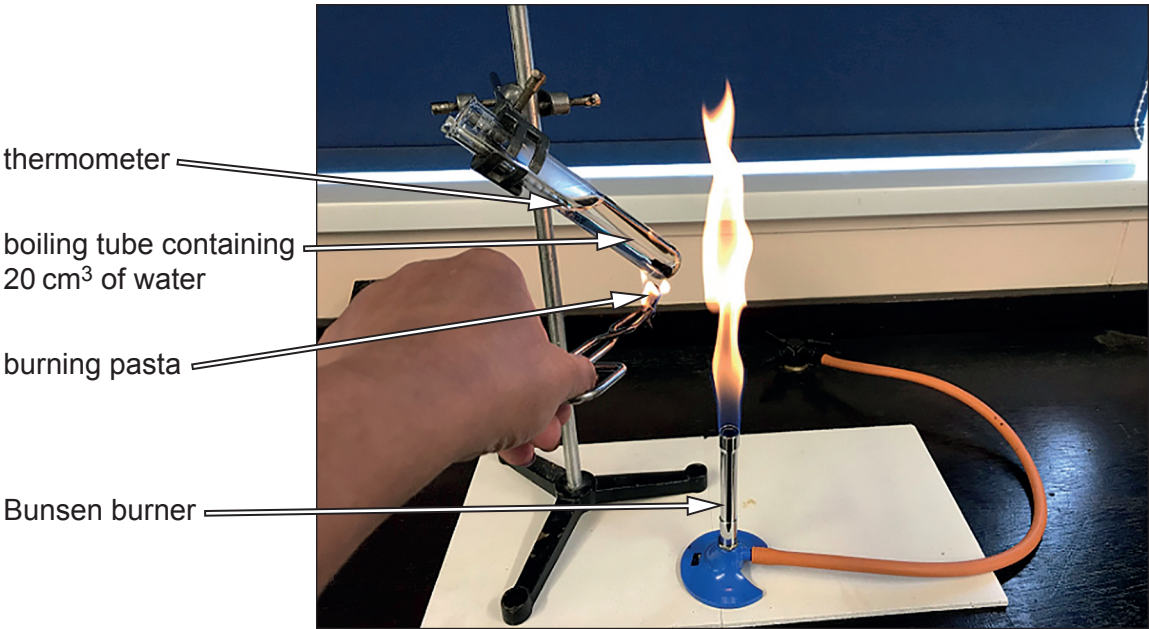
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- (b) Lloyd and Emma carried out an experiment to compare the energy values in **Table 1.1** with values they obtained using the apparatus shown in **Image 1.2**.

Image 1.2



They ignited a 1.6 g piece of dried pasta using the Bunsen burner and immediately held the burning pasta at the base of the boiling tube until it stopped burning. The results Lloyd and Emma obtained are shown in **Table 1.3**.

Table 1.3

Mass of pasta (g)	Initial temperature of water (°C)	Final temperature of water (°C)	Increase in temperature of water (°C)	Energy released per gram of food (kJ)
1.6	14	58	44	

- (i) Use the following formula to calculate the energy released per gram of food (kJ).
Write your answer in Table 1.3. [2]

Energy released per gram (kJ) = $\frac{\text{volume of water (cm}^3\text{)} \times \text{temperature increase (}^\circ\text{C)} \times 0.0042}{\text{mass of pasta sample (g)}}$

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(ii) I. State how the energy content of dried pasta in **Table 1.3** compares to the energy content indicated in **Table 1.1**. You must use numerical data in your answer. [2]

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II. Give **one** reason for the difference between the energy content of dried pasta obtained by Lloyd and Emma, as shown in **Table 1.3** and the energy content indicated in **Table 1.1**. [1]

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(iii) Evaluate the arrangement of the apparatus shown in **Image 1.2** by identifying **one** source of error. [1]

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